FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
CONCEPT RECAPITULATION TEST – IV
PAPER –1
TEST DATE: 31-07-2022

Time Allotted: 3 Hours | **Maximum Marks: 180**

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-B & Section-C**.

Section – A ***(01 – 04, 20 – 23, 39 – 42)*****:** This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A ***(05 –10, 24 – 29, 43 – 48)*****:** This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B ***(11 – 13, 30 – 32, 49 – 51)*****:** This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.**

Section – C ***(14 – 19, 33 – 38, 52 – 57)*****:** This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

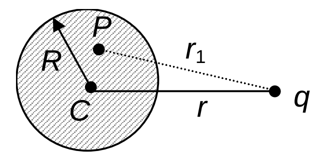
Physics

PART – I

Section – A (Maximum Marks: 12)

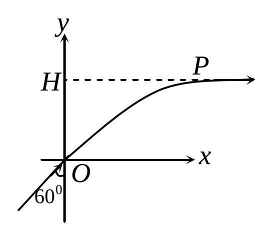
*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

1. A point charge is placed at a distance r from center of a conducting neutral sphere of radius R (r > R). The potential at any point P inside the sphere at a distance r_1 from point charge due to induced charge of the sphere is given by [$k = \frac{1}{4\pi\varepsilon_0}$]



(A) kq/r_1
(B) kq/r
(C) $kq/r - kq/r_1$
(D) $-kq/R$

2. A system of coordinates is drawn in a medium whose refractive index varies as $\mu = \frac{2}{1+y^2}$, where $0 \le y \le 1$. A ray of light is incident at origin at an angle 60^0 with *y* –axis as shown in the figure. At point P ray becomes parallel to x-axis. The value of *H* is



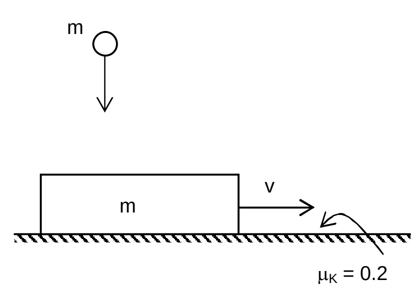
(A) $\left\{\left(\frac{2}{\sqrt{3}}\right)-1\right\}^{1/2}$

(B) $\left\{\frac{2}{\sqrt{3}}\right\}^{1/2}$

(C) $\left\{\left(\sqrt{3}\right)-1\right\}^{1/2}$

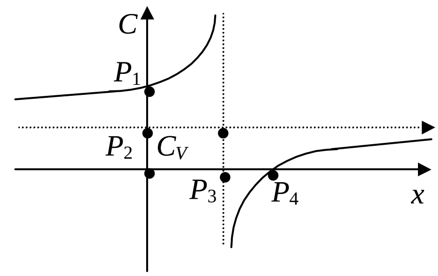
(D) $\left\{\left(\sqrt{3}+1\right)\right\}^{1/2}$

3. A ball of mass m falls vertically from a height h and collides with a block of equal mass m moving horizontally with a velocity v on a surface. The coefficient of kinetic friction between the block and the surface is $\mu_k = 0.2$, while the coefficient of restitution (e) between the ball and the block is 0.5. There is no friction acting between the ball and the block. The velocity of the block just after the collision decreases by (if it is still in motion)



(A) $0.5\sqrt{2gh}$
(B) 0
(C) $0.1\sqrt{2gh}$
(D) $0.3\sqrt{2gh}$

4. The following graph gives the variation of molar heat capacity C a function of x where x is given by the expression PV^x = constant. If gas is monoatomic what are the coordinates of P_1 and P_4



(A) $\left(0, \frac{5}{2}R\right)$ & $\left(\frac{5}{3}, 0\right)$

(B) $\left(0, \frac{3}{2}R\right)$ & $\left(\frac{7}{5}, 0\right)$

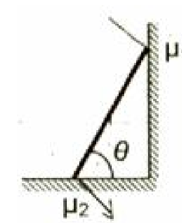
(C) $\left(0, \frac{5}{2}R\right)$ & $\left(\frac{7}{5}, 0\right)$

(D) $\left(0, \frac{7}{5}R\right)$ & $\left(\frac{5}{3}, 0\right)$

Section – A (Maximum Marks: 24)

This section contains ***SIX (06)*** *questions. Each question has* ***FOUR*** *options (A), (B), (C) and (D).* ***ONE OR MOER THAN ONE*** *of these four option(s) is (are) correct answer(s).*

5. In figure, a ladder of mass m is shown leaning against a wall. It is in static equilibrium making an angle θ with the horizontal floor. The coefficient of friction between the wall and the ladder is μ_1 and that between the floor and the ladder is μ_2. The normal reaction of the wall on the ladder is N_1 and that of the floor is N_2. If the ladder is about to slip, then



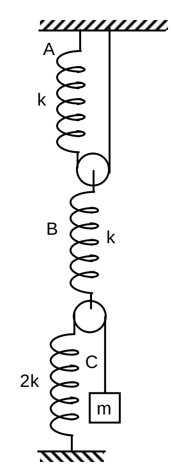
(A) $\mu_1 = 0 \;\; \mu_2 \neq 0$ and $N_2 \tan\theta = \frac{mg}{2}$

(B) $\mu_1 \neq 0 \;\; \mu_2 = 0$ and $N_1 \tan\theta = \frac{mg}{2}$

(C) $\mu_1 \neq 0 \;\; \mu_2 \neq 0$ and $N_2 = \frac{mg}{1+\mu_1\mu_2}$

(D) $\mu_1 = 0 \;\; \mu_2 \neq 0$ and $N_1 \tan\theta = \frac{mg}{2}$

6. In the given figure, the block is attached with a system of three ideal springs A, B, C. The block is displaced by a small distance x from its equilibrium position vertically downwards and released. T represents the time period of small vertical oscillations of the block. Then (pulleys are ideal). Choose the correct alternative(s).



(A) $T = 2\pi\sqrt{\frac{11m}{2k}}$

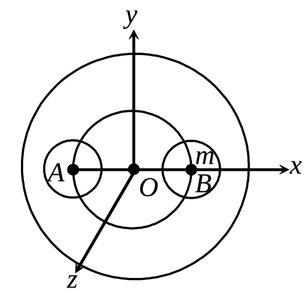
(B) the deformation of the spring A is (2/11) times the displacement of the block.

(C) the deformation of the spring C is (1/11) times the displacement of the block.

(D) the deformation of the spring B is (4/11) times the displacement of the block.

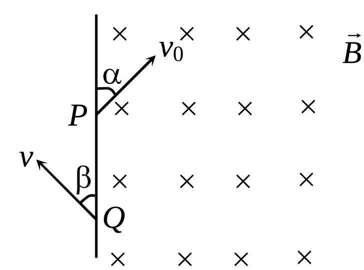
7. One mole of monoatomic gas expands with temperature according to the relation $V = KT^{2/3}$ where 'k' is constant. Then choose the correct alternative(s).
(A) Work done to change the temperature by 30°C is 20 R.
(B) Change in internal energy when temperature is changed by 20°C is 30 R.
(C) If volume of gas is changed by 2% the temperature will change by 3%.
(D) If temperature of gas is changed by 2% the pressure will change by 6%.

8. A solid sphere of uniform density and radius 4 units is located with its centre at the origin *O* of co-ordinates. Two spheres of equal radii 1 units, with their centres at *A*(–2,0,0) and *B* (2, 0 ,0) respectively, are taken out of the solid leaving behind spherical cavities as shown in figure. Then



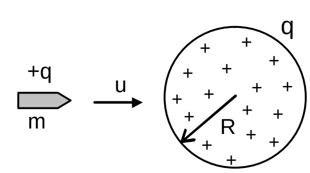
(A) the gravitational field due to this object at the origin is zero.
(B) the gravitational field at the point *B*(2, 0, 0)is zero
(C) the gravitational potential is same at all points on the circle $y^2 + z^2 = 36$
(D) the gravitational potential is same at all points on the circle $y^2 + z^2 = 4$

9. A particle of charge $-q$ and mass m enters a uniform magnetic field $\vec{B}$ (perpendicular to paper inwards) at P with a velocity v_0 at an angle α and leaves the field at Q with velocity v at angle β as shown in the figure. Then



(A) $\alpha = \beta$
(B) $v = v_0$
(C) $PQ = \frac{2mv_0 \sin\alpha}{Bq}$
(D) particle remains in the field for time $t = \frac{2m(\pi - \alpha)}{Bq}$

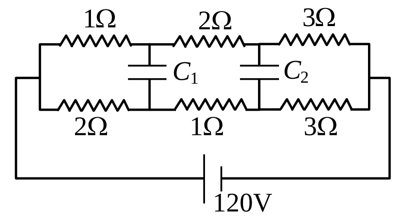
10. Which of these options are correct?



(A) A bullet of mass m and charge q is fired towards a solid uniformly charged sphere of radius R and total charge +q. If it strikes the surface of sphere with speed u. The minimum speed u so that it can penetrate through the sphere. (Neglect all resistance forces or friction acting on bullet except electrostatic forces) is $\frac{q}{\sqrt{4\pi\varepsilon_0 mR}}$
(B) In the above diagram the minimum speed u so that it can penetrate through the sphere. (Neglect all resistance forces or friction acting on bullet except electrostatic forces) is $\frac{q}{\sqrt{2\pi\varepsilon_0 mR}}$

(C) In the circuit shown in figure $C_1 = C_2 = 2\mu F$. Then charge stored in capacitor C_2 is zero

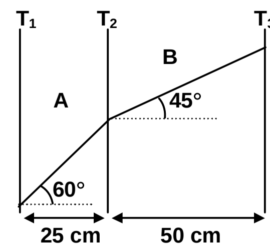
(D) In the above circuit charged stored in capacitor C_1 is 40 μC.



Section – B (Maximum Marks: 12)

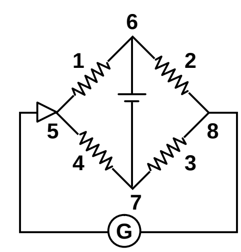
*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

11. Two conductors A and B each of cross section area 5 cm^2 are connected in series. Variation of temperature (in °C) along the length (in cm) is as shown in the figure. If thermal conductivity of A is $2\sqrt{3}$ J/m-sec-°C and the heat current in the conductors is 0.173 J/s. Find thermal conductivity of B (in J/m-sec-°C upto one decimal place.)



12. If voltage V = (100 ± 5) V and current I = (10 ± 0.2) A, the percentage error in resistance R is

13. Four wires of equal length 8 m are arranged as shown in the figure. Wires, 2, 3 and 4 are of equal cross sectional area and wire 1 is of half the cross section of these wires. By how much distance pointer at point 5 must be moved to get null point?



Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 14 and 15

Question Stem

The latent heat of fusion of ice is 80 cal/gram and latent heat of steam is 540 cal/gram. Change of state occurs only at melting point or boiling point of the substance. There is no change in temperature during the entire change of state. For rise in temperature (ΔT), heat required, ΔQ = c.m ΔT, where c is specific heat of substance

14. Find Heat required to melt 10g of ice at 0^0C to water at 0^0C (in Kcal).

15. Find Heat required to convert 1gm of ice at -5^0C to water at 0^0C in Cal. (sp. Heat of ice = 0.5 cal/g/^{0}C)

Question Stem for Question Nos. 16 and 17

Question Stem

A body of mass 1kg is suspended from a massless spring having force constant 600 N/m. Another body of mass 0.5 kg moving vertically upwards hits the suspended body with a velocity 3 m/sec and gets embedded in it (ignore gravity).

16. The frequency (in Hz) of oscillation is _______.

17. The amplitude of oscillation (in m)

Question Stem for Question Nos. 18 and 19

Question Stem

The equation of a progressive wave travelling along positive x axis is y(x,t) = $r\sin\frac{2\pi}{\lambda}(vt - x + \phi_0)$, where ϕ_0 is initial phase and other symbols have their usual meaning. When the wave is travelling along negative x – axis, vt and x have same sign. It a homogeneous medium the velocity of wave v is constant and its acceleration is zero. The particle velocity, dy/dt keeps on changing

18. In the equation $y = 0.25\cos\left[\frac{2\pi t}{5} - \frac{2\pi x}{3} + \frac{\pi}{4}\right]$, where y, x are in metre and t in seconds then find the amplitude (in m) of the wave.

19. The frequency (in Hz) of the wave is

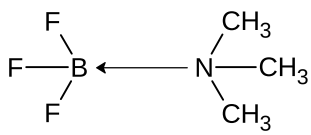
Chemistry

PART – II

Section – A (Maximum Marks: 12)

This section contains ***FOUR (04)*** *questions. Each question has* ***FOUR*** *options.* ***ONLY ONE*** *of these four options is the correct answer.*

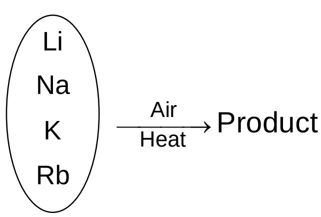
20. Choose the correct statement regarding the following molecule



(A) $\angle FBF$ bond angle is 120^o
(B) Nitrogen undergoes sp^2 hybridization
(C) $p\pi$ - $p\pi$ back bond between B & F in the molecule is more dominant than that in isolated BF_3 molecule
(D) All central atoms have the same type of hybridization

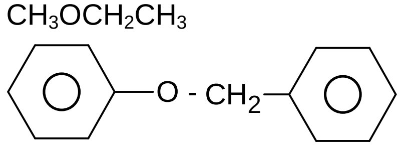
21. The pH of 0.01 M NaCl is identical to that of
(A) 0.01 M Na_2CO_3
(B) 0.01 M NaOH
(C) 0.01 M KNO_3
(D) 0.01 M $KHCO_3$

22.

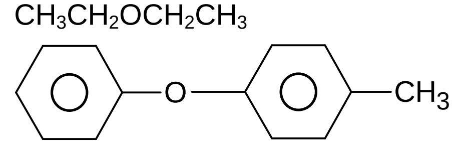


Which product is not formed in the above reaction?
(A) Na_2O_2
(B) Li_3N
(C) KH
(D) RbO_2

23. Which ether forms the maximum number of product(s) with HI?
(A) $CH_3OCH_2CH_3$
(B) $CH_3CH_2OCH_2CH_3$
(C)



(D)



Section – A (Maximum Marks: 24)

This section contains ***SIX (06)*** *questions. Each question has* ***FOUR*** *options (A), (B), (C) and (D).* ***ONE OR MOER THAN ONE*** *of these four option(s) is (are) correct answer(s).*

24. Which of the following solution(s) form buffer if they are taken in 2 : 1 molar ratio?
(A) CH_3COOH and CH_3COONa
(B) NH_4OH and NaCl
(C) NH_4OH and NH_4Cl
(D) $NaHCO_3$ and Na_2CO_3

25. Which of the following compound(s) on catalytic hydrolysis forms carbonyl compound(s)?
(A) $CH_3CH = CH_2$
(B) $CH_3C \equiv CH$
(C) $CH_3C \equiv CCH_3$
(D) $HC \equiv CH$

26. Which of the following compound(s) contain(s) hydrogen atoms?
(A) Silicate anions
(B) Silicones
(C) Silanes
(D) Selenes

27. Which of the following complex(es) is/are soluble in water?
(A) $Na_2[Zn(CN)_4]$
(B) $[Co(NH_3)_3Cl_3]$
(C) $[Cr(H_2O)_6]Br_3$
(D) $H_2[SiF_6]$

28. $FeCl_3 + K_4\left[Fe\left(CN\right)_6\right] \longrightarrow \text{Precipitate} \downarrow$
The correct statement(s) regarding the precipitate in above reaction is/are
(A) it is blue in colour
(B) one mole of the precipitate can consume 12 moles of NaOH
(C) the maximum van't Hoff factor of the precipitate can be seven
(D) the precipitate becomes soluble in hot water

29. $Zn(s) \mid Zn^{2+}(C_1) \parallel H^+(C_2) \mid H_2(1\text{ atm}), Pt(s)$
Under which conditions, E_{Cell} will be equal to E^o_{Cell}?
(A) $C_1 = C_2 = 1$ M
(B) $C_2 = \sqrt{C_1}$
(C) $C_1 = C_2 = 0.1$ M
(D) $C_1 = \sqrt{C_2}$

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

30. $CH_3COCH_3 \xrightarrow[\text{Heat}]{\text{Conc.}H_2SO_4} \text{Product}\left(P\right)$
If the number of hydrogen atom(s) present in (P) is 'X', what is the value of $\frac{X}{2}$?

31. The ionization constants Ka_1 and Ka_2 of a weak acid H_2A are 10^{-6} and 10^{-10} respectively. What is the pH of 0.1 M solution of NaHA which is formed in the following reaction?
$NaOH + H_2A \rightleftharpoons NaHA + H_2O$

32. $$S + O_2 \longrightarrow (A) \xrightarrow[\Delta]{O_2/Pt} (B) \xrightarrow{H_2SO_4} (C)$$

How many covalent bond(s) is/are present in (C)?

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 33 and 34

Question Stem

An electrochemical cell is set up by combining the Zn electrode with hydrogen electrode. Zinc electrode behaves as anode and the hydrogen electrodes as cathode. The concentration of Zn^{2+} ions in the electrolyte is 0.01 M and that of H^+ ions in the electrolyte is 'C' M. The cell is represented as:

$Zn(s) \mid Zn^{2+}(0.01\ M) \parallel H^+(CM) \mid H_2(1\ bar), Pt(s)$

x = pH of hydrogen electrode when $E_{cell} = E^o_{cell}$

y = The potential of the hydrogen electrode in volt unit when $E_{cell} = E^o_{cell}$

33. The value of x is

34. The value of |y| upto two decimal point is

Question Stem for Question Nos. 35 and 36

Question Stem

$$P_4 + O_2\,(\text{limited}) \longrightarrow (A) \xrightarrow{H_2O} (B) \xrightarrow{\text{Heat}} (D) + (C)$$

Compound(D) is obtained by reaction between Ca_3P_2 and HCl.

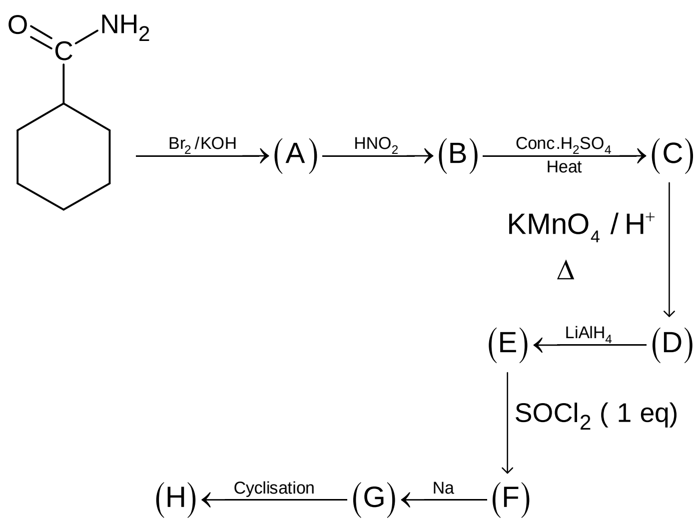
Oxidation number of phosphorus in compound(B) is +x. The total number of covalent bonds present in (C) is 'y'.

35. What is the value of x?

36. What is the value of y?

Question Stem for Question Nos. 37 and 38

Question Stem



x = Number of oxygen atoms in compound(D)
y = Number of hydrogen atoms in compound(H)

37. The value of x is

38. The value of y is

Mathematics

PART – III

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

39. If f(x) be an invertible function with $f'(x)f''(x) > 0 \forall x \in R$, then $\frac{d^2f^{-1}(x)}{dx^2}$, where $f^{-1}(x)$ is the inverse function of f(x) is
(A) positive $\forall x \in R$
(B) negative $\forall x \in R$
(C) data insufficient
(D) $\frac{d^2f^{-1}(x)}{dx^2}$ will have sign opp. to $\frac{d^2f^{-1}(x)}{dx^2}$

40. For $\alpha \in R$, the range of the function $f(\alpha) = \int_{\tan^{-1}\alpha}^{\cot^{-1}\alpha}\left(\frac{\tan x}{\tan x + \cot x}\right)dx$ is equal to
(A) $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$
(B) $(0, \pi)$
(C) $\left(\frac{-\pi}{4}, \frac{3\pi}{4}\right)$
(D) $\left(0, \frac{\pi}{2}\right)$

41. If $1, \alpha_1, \alpha_2, \alpha_3, \alpha_4$ are the fifth roots of unity then $\sum_{r=1}^{4}\frac{1}{2-\alpha_r} =$
(A) $\frac{51}{31}$
(B) $\frac{49}{31}$
(C) $\frac{25}{32}$
(D) $\frac{25}{16}$

42. If α, β and γ are three real roots of the equation $x^3 - 6x^2 + 5x - 1 = 0$, then the value of $\alpha^4 + \beta^4 + \gamma^4$ is:
(A) 250
(B) 650
(C) 150
(D) 950

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

43. Let S = {1, 2, 3, 4,……..,n} and f_n be the number of those subsets of S which do not contain consecutive elements of S, then

(A) $f_n = \frac{n(n-1)(n-2)}{6}$

(B) $f_n = 2f_{n-1}$

(C) $f_n = f_{n-1} + f_{n-2}$

(D) $f_4 = 8$

44. Let $f(x) = x + \frac{8}{9}x^3 - \tan x$, where $0 < x < \frac{\pi}{6}$, then

(A) $f'(x) > 0$

(B) $f''(x) > 0$

(C) $f'''(x) > 0$

(D) $f(x) > 0$

45. Given that x_1, x_3 are roots of the equation $ax^2 - 4x + 1 = 0$ and x_2, x_4 are roots of the equation $bx^2 - 6x + 1 = 0$. If x_1, x_2, x_3, x_4 are in harmonic progression, then

(A) $3a - b = 1$

(B) $a^2 + b^2 = 73$

(C) $2a < 3b$

(D) $\frac{1}{a} > \frac{1}{b}$

46. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three vectors having magnitudes 1, 1 and 2 respectively. If $\vec{a} \times (\vec{a} \times \vec{c}) + \vec{b} = 0$, then

(A) $\vec{b}$ is in the plane of $\vec{a}$ and $\vec{c}$

(B) $\vec{b}$ is parallel to $\vec{a} \times \vec{c}$

(C) angle between $\vec{a}$ and $\vec{c}$ is 30^o

(D) angle between $\vec{a}$ and $\vec{c}$ is 150^o

47. If A and B are two orthogonal matrices of order n and $\det(A) + \det(B) = 0$, then which of the following must be correct

(A) $\det(A + B) = \det(A) + \det(B)$

(B) $\det(A + B) = 0$

(C) A + B = 0

(D) None of above are always correct

48. Let the complex number $z = x + iy$ (where $x, y \in R$) satisfy
$\cot^{-1}\left(\log_3 |2z+1|\right) > \cot^{-1}\left(\log_3 |2z-1|\right)$, then x can be
(A) –1
(B) –2
(C) 1
(D) 2

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

49. An urn contain 3 red balls and n white balls. Mr. A draws two balls together from the urn. The probability that they have the same colour is 1/2. Mr. B draws one ball from the urn, notes its colour and replaces it. He then draws a second ball from the urn and finds that both balls have the same colour is $\frac{5}{8}$. The possible value of n is_________

50. Consider the system of equations $ax + y = b, bx + y = a, ax + by = ab$ where $a, b \in \{0, 1, 2, 3, 4\}$. Find the number of ordered pairs (a, b) for which system is consistent.

51. Let $f(x)$ be a differentiable function in $[-1, \infty)$ and $f(0) = 1$ such that
$\lim\limits_{t \to x+1} \frac{t^2 f(x+1) - (x+1)^2 f(t)}{f(t) - f(x+1)} = 1$. Then value of $\lim\limits_{x \to 1} \frac{\ln(f(x)) - \ell n 2}{x-1} =$

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 52 and 53

Question Stem

Suppose a. b, c are three distinct real numbers and p (x) is a real quadratic polynomial such that

$$\begin{bmatrix} 4a^2 & 4a & 1 \\ 4b^2 & 4b & 1 \\ 4c^2 & 4c & 1 \end{bmatrix} \begin{bmatrix} p(-1) \\ p(1) \\ p(1) \end{bmatrix} = \begin{bmatrix} 3a^2 + 3a \\ 3b^2 + 3b \\ 3c^2 + 3c \end{bmatrix}$$

52. The absolute value of x – coordinates (s) of the point of intersection of $y = f(x)$ with the x – axis

53. Area bounded by $y = \frac{3}{2} f(x)$ and the x – axis

Question Stem for Question Nos. 54 and 55

Question Stem

A plane π passes through the point (1,1, 1) and is parallel to the vectors $\vec{a}$ = (–1, 1, 0) and $\vec{b} = (1, 0, -1)$ π meets the axis in A, B and C forming the tetrahedron OABC where O is origin.

54. Volume of the tetrahedron OABC in cubic units

55. Distance of the points $\left(3\sqrt{3}/2, 3\sqrt{3}, 3\right)$ form the plane π in units

Question Stem for Question Nos. 56 and 57

Question Stem

If the function $f(x) = ax\,e^{-bx}$ has a local maximum at the point (2, 10) then

56. Then the value of $\frac{A}{e}$ is

57. The value of B is